

Mxolisi Khumalo

Gauteng | 0716147319 | mxolisikhumalo611@gmail.com | [Portfolio](#) | [LinkedIn](#) | [GitLab](#)

PROFESSIONAL SUMMARY

Software Developer & Engineer with strong foundations in Java, Python, and Cloud Computing, experienced in building concurrent systems, mentoring developers, and delivering production-ready projects using Agile practices. Able to design scalable, thread safe applications, deploy cloud solutions, and collaborate effectively in technical teams. Passionate about solving real-world problems through efficient high performing software.

CORE SKILLS

Programming Languages: Python, Java, JavaScript, C#, SQL, HTML/CSS, Assembly, R
Software Engineering: OOP, Data Structures & Algorithms, Multithreading & Concurrency, Parallel Computing, Debugging, Unit Testing

Cloud & Frameworks: Microsoft Azure, AWS, Cloud Infrastructure Fundamentals, Network Security Basics, Angular, React, Node

Tools & Platforms: Git, Docker, VS Code, IntelliJ IDEA, Eclipse, JGrasp, RStudio, QtSpim

Methodologies: Agile/Scrum, Pair Programming, Code Reviews, version control

Soft Skills: Technical Mentorship, Team Leadership, Communication, Problem Solving

FEATURED PROJECTS

Self-Storage Platform(team project)| *AWS, Python, HTML/CSS/JavaScript, Postman*

Developed a serverless self-storage web application allowing customers to browse facilities/units, book instantly, manage rentals and configure security features.

- Built a static SPA-style front end in vanilla HTML/CSS/JS hosted on Amazon S3 and distributed via Amazon CloudFront.
- Implemented backend business logic using AWS Lambda (Python) behind Amazon API Gateway, with data persisted in Amazon DynamoDB.
- Designed async event handling for workflows/notifications using Amazon SQS and EventBridge; implemented email notifications via Amazon SES (sandbox).

Concurrent Thread Simulation | *Java (Swing/AWT), Multithreading, Atomic Variables*

Engineered a real-time simulation modelling resource sharing and thread coordination.

- Implemented thread synchronisation using monitors to eliminate race conditions.
- Designed non-blocking state management with AtomicBoolean and CountdownLatch for pause/resume controls.
- Made deadlock-prevention ensuring continuous agent flow under high concurrency.

Document Signing App | *Angular, TypeScript, Tailwind CSS, PrimeNG, pdf-lib*

Developed a serverless document signing application allowing users to upload PDFs, position signature fields, and sign electronically via draw or type inputs.

- Engineered a drag-and-drop interface using Angular CDK, mapping HTML DOM coordinates to PDF internal points for precise field placement.
- Implemented client-side PDF manipulation using pdf-lib to permanently embed signature images and text, flattening documents for secure download.
- Integrated pdf-viewer and signature_pad within a responsive Angular standalone architecture to deliver a seamless signing experience without backend.

PROFESSIONAL EXPERIENCE

Technology Associate | ActiveOps | Johannesburg, Gauteng | Mar 2027 - Present

- Contribute to feature delivery on an enterprise SaaS platform within a multi-team Agile environment, working across backend services, APIs, and data layers.
- Translate business requirements into user stories, acceptance criteria, and API specifications, bridging development and business analysis across delivery squads.
- Analyse and work within a multi-tenant, role-based enterprise architecture, resolving cross-team dependencies and interface contracts ahead of build.

Technical Mentor | WeThinkCode_ | Johannesburg, Gauteng | Sep 2025 - Jan 2026

- Mentored 15+ software engineering students in Python, improving project pass and completion rates by approximately 30%.
- Facilitated pair programming sessions and conducted 50+ structured code reviews improving architecture and code quality.
- Supported debugging, algorithmic thinking, and performance optimisation across multiple teams.

Campus Role Supervisor | WeThinkCode_ | Springs, Gauteng | Oct 2025 - Jan 2026

- Directed operational allocation for 10+ students, ensuring 100% compliance with performance and campus role requirements.
- Managed schedules, attendance tracking, and performance reporting
- Acted as liaison between students, facilitators, and operations management.

Academic Tutor | 123tutors | Cape Town, South Africa | June 2023 - Sep 2024

- Tutored high school and university students in technical and analytical subjects.
- Developed personalized study plans, resulting in improved academic performance.
- Achieved 100% tutored students pass rate.

EDUCATION

WeThinkCode_ | Johannesburg

National Diploma: Software Development (NQF Level 6) | Sep 2024 - Dec 2025

Focus: Software Developing & Engineering, Cloud Computing | *Certificate Pending*

University of Cape Town | Cape Town

Undergraduate(Bsc) Coursework: Information Technology | Feb 2020 – Feb 2024

Did four years of studying covering Computer Science, Computer Engineering, Information Systems, and Statistics.

Zimisele Secondary School | Springs

National Senior Certificate: CAPS | Jan 2015 – Dec 2019

Top 10 Matriculant Achiever | Average: 77% | 2 Distinctions

CERTIFICATIONS

- **WeThinkCode_ generative AI**
- **Microsoft Azure Technologies**
- **AWS Cloud Trainee**
- **Business Analysis Foundations**
- **Scrum Master**
- **Software Testing Foundations**